

Unit – V [Laplace Transforms]

Find the Laplace transforms of the following functions.

(a) $\sinh t \cos t$ (b) $2te^{-t} \cos^2 \frac{1}{2}t$ (c) $(t+1)^2 e^{2t}$

(d) $[5t^5 - 2t^4 + 4e^{-3t} - 3\sin 6t + 4\cos 4t]e^{-2t}$ (e) $e^{\frac{1}{2}t} \sin 2t \cos 3t$ (f) $\sinh 3t \cos^2 2t$

(g) $4u(t-\pi) \cos^{2\frac{1}{2}}t$ (h) $\int_0^t \frac{e^{-t} \sin t}{t} dt$ (i) $\int_0^t \frac{e^{-t} \cos t \sin t}{t} dt$ (j) $\int_0^t \frac{e^{-t} \cos t \sin 3t}{t} dt$

(k) $f(t) = \begin{cases} t^2, & 0 \leq t \leq 2 \\ \sin 2t, & 2 \leq t \leq 5 \\ \cosh t, & 5 \leq t \leq 7 \\ e^{-2t}, & t \geq 8 \end{cases}$ (l) $f(t) = \begin{cases} t-1, & 1 < t < 2 \\ 3-t, & 2 < t < 3 \end{cases}$, (m) $e^{-t} \{1 - u(t-2)\}$

Find the inverse Laplace transforms of the following functions.

(i) $\ln \frac{s+a}{s+b}$ (ii) $\cot^{-1} \left(\frac{s}{\pi} \right)$ (iii) $\frac{s}{(s^2-9)^2}$ (iv) $\frac{s}{(s^2+\pi^2)^2}$

(v) $\frac{s^3-7s^2+14s-9}{(s-1)^2(s-2)^3}$ (vi) $\frac{s^2+2s+3}{(s^2+2s+2)(s^2+2s+5)}$ (vii) $\frac{s}{s^4+4a^4}$ (viii) $\frac{\omega}{s^2(s^2+\omega^2)}$

(ix) $\frac{s^2}{s^4+4a^4}$ (x) $\frac{s}{s^4-\omega^4}$ (xi) $\frac{3s+7}{s^2-2s-3}$ (xii) $\frac{s^2+s-2}{s(s+3)(s-2)}$ (xiii) $\frac{s^2-10s+13}{(s-7)(s^2-5s+6)}$

(xiv) $\frac{1+2s}{(s+2)^2(s-1)^2}$ (xv) $\frac{s}{(s-3)(s^2+4)}$ (xvi) $\frac{s}{(s+1)^2(s^2+1)}$ (xvii) $\frac{s^3}{s^4-a^4}$

(xviii) $\frac{3(1-e^{-\pi s})}{(s^2+9)}$ (xix) $\frac{s^2-\pi^2}{(s^2+\pi^2)^2}$ (xx) $\frac{s^3}{s^4+4a^4}$ (xxi) $\frac{1}{s^4+4a^4}$ (xxii) $\frac{1}{s^3-a^3}$

(xxiii) $\frac{s}{s^4+s^2+1}$ (xxiv) $\log \left[\frac{s^2+1}{(s-1)^2} \right]$ (xxv) $s \log \left[\frac{s-1}{s+1} \right]$ (xxvi) $\frac{1}{s(s+1)^3}$ (xxvii) $\cot^{-1} \left[\frac{s+a}{b} \right]$

(xxviii) $\tan^{-1}[s+1]$ (xxix) $\frac{s}{s^4-s^2+1}$ (xxx) $\frac{s}{(s-a)(s^2+\omega^2)}$ (xxxi) $\frac{\omega}{(s^2-\omega^2)(s^2+\omega^2)}$

3.	Using Laplace's Transforms solve the following initial value problems. (i) $y'' + 2y' + 2y = 5\sin t, y(0) = y'(0) = 0$ (ii) $y' + 3y = 10\sin t, y(0) = 0$ (iii) $y'' + 2y' + y = e^{-t}, y(0) = -1, y'(0) = 1$ (iv) $y'' + 2y' + 2y = 5\sin t, y(0) = y'(0) = 0$ (v) $y'' + 9y = r(t), y(0) = 1, y'(0) = 4$, where $r(t) = \begin{cases} \sin t & \text{if } 0 < t < \pi \\ 0 & \text{if } t > \pi \end{cases}$ (vi) $y'' + 3y' + 2y = r(t), y(0) = 0, y'(0) = 0$, where $r(t) = \begin{cases} 4t & \text{if } 0 < t < 1 \\ 8 & \text{if } t > 1 \end{cases}$ (vii) $y'' + 2y' - 3y = 8e^{-t} + \delta\left(t - \frac{1}{2}\right), y(0) = 3, y'(0) = -5$ (viii) $y'' - 5y' + 6y = u(t - 1) + \delta(t - 2), y(0) = 0, y'(0) = 1$
4.	Using convolution theorem solve the following initial value problems. (i) $y'' + y = \sin t, y(0) = 0, y'(0) = 0$ (ii) $y'' + y = t, y(0) = 0, y'(0) = 0$ (iii) $y'' + y = 3\cos 2t, y(0) = 0, y'(0) = 0$ (iv) $y'' + y = r(t), r(t) = t$ if $1 < t < 2$ and 0 otherwise, $y(0) = 0, y'(0) = 0$
5.	Using Laplace transform determine current in the LR circuit with $L = 1\text{henry}$, $R = 100\text{ohms}$, initial current is 0, and the voltage given by $v(t) = \begin{cases} t & \text{if } 0 < t < 4\pi \\ 0 & \text{if } t > 4\pi \end{cases}$
6.	Using Laplace transform determine current in the LC circuit with $L = 1\text{henry}$, $C = 1\text{Farad}$, 0 initial current and charge, and the voltage given by $v(t) = \begin{cases} t & \text{if } 0 < t < 1 \\ 1 & \text{if } t > 1 \end{cases}$
7.	Using Laplace transform determine current in the RC circuit with $C = 0.1\text{Farad}$, $R = 100\text{ohms}$, 0 initial current and charge, and the voltage given by $v(t) = \begin{cases} 0 & \text{if } t < 2 \\ e^{-t} & \text{if } t > 2 \end{cases}$
8.	Using Laplace transforms solve the following integral equations: (i) $y(t) + \int_0^t y(\tau)(t - \tau)d\tau = 1$ (ii) $y(t) = t + \int_0^t y(\tau)\sin(t - \tau)d\tau$ (iii) $y(t) = te^t - 2e^t \int_0^t e^{-\tau}y(t - \tau)d\tau$ (iv) $f(t) = t + e^{-2t} + \int_0^t f(\tau)e^{2(t-\tau)}d\tau$
9.	State and prove Convolution Theorem of Laplace Transform
10.	State and prove Second Shifting Theorem of Laplace Transforms
11.	Prove that if $L\{f(t)\} = F(s)$ then $L\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(\bar{s})d\bar{s}$
<u>SHORT QUESTIONS</u>	
1.	State under what conditions Laplace Transform of a function exists.
2.	State and prove the First Shifting Theorem of Laplace Transforms
3.	Prove that if $L\{f(t)\} = F(s)$ then for any constant a , $L\{f(at)\} = \frac{1}{a}F\left(\frac{s}{a}\right)$
4.	Prove that if $L\{f(t)\} = F(s)$ then $L\{f'(t)\} = sF(s) - f(0)$
5.	Prove that if $L\{f(t)\} = F(s)$ then $L\left\{\int_0^t f(\tau)d\tau\right\} = \frac{F(s)}{s}$

6.	Define unit step / Heaviside function
7.	Find the Laplace Transform of Unit Step function
8.	Define Dirac Delta Function.
9.	Find the Laplace Transform of Dirac Delta function.
10.	Prove that if $L\{f(t)\} = F(s)$ then $L\{tf(t)\} = -F'(s)$.
11.	State Convolution Theorem of two functions.
12.	Evaluate the convolution $t * e^t$
13.	Evaluate the convolution $e^{at} * e^{bt}$ ($a \neq b$)
14.	Evaluate the convolution $t * \sin t$
15.	Show that convolution of two functions exhibit commutative property.
16.	State second shifting theorem of Laplace transforms
17.	Can a discontinuous function have a Laplace transform? Give reason for your answer.
18.	Give an example to illustrate that there may be two or more functions with same Laplace transform.
19.	Find the Laplace transform of $f(t) = \begin{cases} t & \text{if } 0 \leq t \leq 1 \\ 2-t & \text{if } 1 \leq t \leq 2 \\ 0 & \text{if } t > 2 \end{cases}$
20.	Find the Laplace transforms of $L\{t^{-1/2}\}$ and $L\{t^{1/2}\}$
21.	Find the Laplace transforms of the following functions (i) $\sin 2t \sin 3t$ (ii) $\sin^3 2t$ (iii) $\cos^2 2t$ (iv) $f(t) = \begin{cases} \frac{t}{k}, & 0 < t < k \\ 1, & t > k \end{cases}$.
22.	From the definition find the Laplace transform of $1 + \cos 2t$.
23.	Using Laplace transform evaluate the following integrals (i) $\int_0^\infty e^{-2t} \sin 4t dt$ (ii) $\int_0^\infty e^{-3t} \cosh 3t dt$ (iii) $\int_0^\infty e^{-5t} t^6 dt$ (iv) $\int_0^\infty e^{-t} t^{-\frac{1}{2}} dt$ (v) $\int_0^\infty e^{-t} \sin t \cos 3t dt$
25.	Find the inverse Laplace transforms of the following. (i) $\frac{s}{L^2 s^2 + n^2 \pi^2}$ (ii) $\frac{2s^3}{s^4 - 1}$ (iii) $\frac{-s - 10}{s^2 - s - 2}$ (iv) $\frac{6}{(s-4)(s+2)}$ (v) $\frac{s}{(s+1)^2}$.
26.	Find the Laplace transforms of the following functions. (i) te^t (ii) $t^2 \sin 3t$ (iii) $te^{-t} \cos t$ (iv) 100^{5t} (v) $3e^{-2t} \cos^3 t$ (vi) $(t-1)^3 e^{2t}$ (vii) a^{2t} .
27.	Find the inverse Laplace transforms of the following. i) $\frac{0.1s + 0.9}{s^2 + 3.24}$ (ii) $\frac{1}{(s + \sqrt{2})(s - \sqrt{3})}$ (iii) $\frac{s^4 + 6s - 18}{s^5 - 3s^4}$ (iv) $\frac{2}{s^2 + s + \frac{1}{2}}$
28.	Evaluate the convolution $u_5(t) * \cos 2t$